# BUS105

**Examination - July Semester 2023**

## Statistics

**Thursday, 16 November 2023 10:00 am - 12:00 pm**

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**Time allowed: 2 hours**

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**INSTRUCTIONS TO STUDENTS:**

1. This examination contains **4** questions and comprises **8** printed pages (including cover page).
2. You must answer **4** questions.
3. All answers must be written in the answer book.
4. This is **Open-book** examination.

**At the end of the examination.**
Please ensure that you have written your examination number on each answer book used.

**Failure to do so will mean that your work cannot be identified.**
If you have used more than one answer book, please tie them together with the string provided.

**THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.**

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(Full marks: 100)

***You must answer ALL the questions.*** **(Total 100 marks)**

## Question 1

During the Covid-19 pandemic, all physical courses have been converted to online delivery. This inevitably gives rise to the concern on the impact of students' academic performance with online lessons. In particular, Statistics is a quantitative subject that may face more challenges when delivers online. To examine the impact, Mr Tan, an educationist in SUSS, collected the overall scores for the Statistics course of 50 students in January 2019 semester (before Covid-19) and 45 students in January 2022 semester (during Covid-19). A partial listing of the scores is as follows:

***Statistics Overall Scores in January 2019:***

23, 30, 30, 31, . . . . , 81, 87, 88

***Statistics Overall Scores in January 2022:***

6, 9, 12, 15, . . . . , 87, 89, 98

***Table 1.1*** and ***Figure 1.1*** are some useful information.

| ***Statistics Overall Scores*** | | |
|---|---|---|
| | *2019 Jan* | *2022 Jan* |
| Mean | 56.90 | 50.40 |
| Median | 56.50 | 55.00 |
| Standard Deviation | 15.27 | 19.49 |
| Interquartile Range | 20.00 | 18.00 |
| Count | 50 | 45 |

Table 1.1: Summary statistics of overall scores for 2019 & 2022 January semester

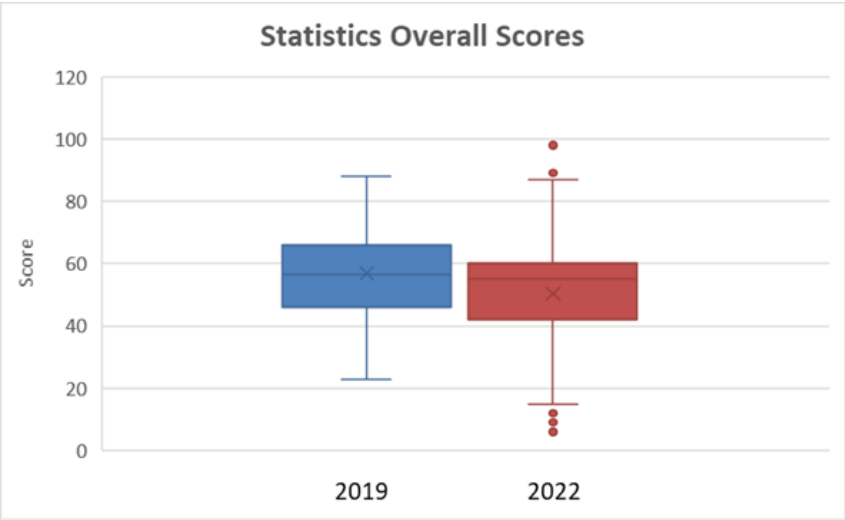

Figure 1.1: Boxplot of Statistics overall scores for 2019 & 2022 January semester

**Question 1a**

Describe how the mean and standard deviation of the Statistics Overall Scores for 2019 Jan were computed by expressing the calculation explicitly. Explain how to choose between mean and median as well as between standard deviation and interquartile range.

(10 marks)

**Question 1b**

Interpret the results presented and report these as an executive summary. You should assist Mr Tan to draw a preliminary conclusion based on your observation.

(15 marks)

## Question 2

### Question 2a

A random survey was conducted on SUSS students to determine students' preference for online lessons. Responses from 200 full-time and 600 part-time students were collected. Of which, 450 part-time students prefer online lessons while 110 full-time students prefer online lessons.

(i) Construct a contingency table based on the information given.

(ii) Identify the probability that a randomly selected student is in a part-time programme and prefers online lessons.

(iii) Identify the probability that a randomly selected student is in a full-time programme or prefer online lessons?

(iv) What is the probability that a randomly selected student is a part-time student given that the student prefers online lessons?

(12 marks)

### Question 2b

The starting salary of a university fresh graduate in Singapore is normally distributed with a mean of $2,500 and standard deviation of $488.

(i) If a fresh graduate is randomly selected, identify the probability that the starting salary is between $1,900 and $2,800.

(ii) If 20 fresh graduates are randomly selected, report the probability that their mean starting salary is greater than $2,750.

(13 marks)

## Question 3

The Head of Business Department in SUSS would like to find out whether there is a difference in motivation to learn Statistics between male and female students. 30 male students and 30 female students were randomly selected to complete a survey on motivation. The total score from the survey (out of 100) reflects a student's motivation to learn Statistics. A higher score means students is more motivated. The summary statistics for the two sets of data are given in ***Table 3.1***, and the test result is given in ***Table 3.2***.

| *Motivation Scores* | | |
|---|---|---|
| | *Male* | *Female* |
| Mean | 60.73 | 53.03 |
| Standard Error | 2.33 | 3.61 |
| Median | 67.50 | 52.00 |
| Mode | 70.00 | 68.00 |
| Standard Deviation | 12.75 | 19.79 |
| Sample Variance | 162.48 | 391.83 |
| Kurtosis | -0.83 | -1.44 |
| Skewness | -0.76 | -0.27 |
| Range | 44 | 64 |
| Minimum | 35 | 15 |
| Maximum | 79 | 79 |
| Sum | 1822 | 1591 |
| Count | 30 | 30 |
| Confidence Level(95.0%) | 4.76 | 7.39 |

Table 3.1: Summary statistics of students' motivation to learn Statistics

| | *Male* | *Female* |
|---|---|---|
| Mean | 60.73333 | 53.03333 |
| Variance | 162.4782 | 391.8264 |
| Observations | 30 | 30 |
| Pooled Variance | 277.1523 | |
| Hypothesized Mean Difference | 0 | |
| df | 58 | |
| t Stat | 1.791336 | |
| P(T<=t) one-tail | 0.039229 | |
| t Critical one-tail | 1.671553 | |
| P(T<=t) two-tail | 0.078457 | |
| t Critical two-tail | 2.001717 | |

Table 3.2: Result of an appropriate hypothesis test

**Question 3a**

Describe the 95% confidence intervals for the means of the motivation scores of male and female students respectively. Based on the confidence intervals, discuss whether we can conclude that the means of the motivation scores for both genders are different.

(8 marks)

**Question 3b**

Apply an appropriate hypothesis test at the 5% significant level to determine whether the means of the motivation scores of both genders are different.

(10 marks)

**Question 3c**

Compare the results from ***Part (a) and (b)***. Should the conclusion be different or similar? Please provide your justification.

(7 marks)

**Question 4**

Mr Tan is a dietician in SUSS. He is researching on factors that affect weights of students. He recorded the following information of 100 students from SUSS:

WEIGHT: Weight of students (kg)
AGE: Age of students (years)
SODIUM: Average daily sodium intake (mg/day)
SMOKING: 1 means the student is smoking
0 means the student is not smoking
Gender: 1 means the student is male
0 means the student is female

The WEIGHT is the dependent variable and the other variables are independent variables. Multiple linear regression is used for the analysis, and the regression output is shown in ***Table 4.1***.

| *Regression Statistics* | |
|---|---|
| Multiple R | 0.76575535 |
| R Square | 0.58638125 |
| Adjusted R Square | 0.57738678 |
| Standard Error | 6.33393659 |
| Observations | 100 |

ANOVA

| | *df* | *SS* | *MS* | *F* | *Significance F* |
|---|---|---|---|---|---|
| Regression | 4 | 14030.23089 | 3507.558 | 87.42938 | 5.51658E-31 |
| Residual | 95 | 3811.281509 | 40.11875 | | |
| Total | 99 | 17841.5124 | | | |

| | *Coefficients* | *Standard Error* | *t Stat* | *P-value* |
|---|---|---|---|---|
| Intercept | -72.421821 | 16.9552025 | -4.27136 | 4.61E-05 |
| AGE | 0.27757555 | 0.049932882 | 5.558973 | 2.48E-07 |
| SODIUM | 0.05162473 | 0.007631839 | 6.764389 | 1.09E-09 |
| SMOKING | 1.89770693 | 1.274635152 | 1.488824 | 0.139846 |
| GENDER | 13.8620059 | 1.418026548 | 9.775562 | 4.98E-16 |

Table 4.1: Statistical report for multiple linear regression analysis

**Question 4a**

Describe the relationship between WEIGHT and the ***four (4)*** independent variables by writing down the linear equation. Interpret the coefficients to show how each of the independent variables impacts weight.

(5 marks)

**Question 4b**

Mr Tan wants to estimate the weight of a male student who is 30 years old with a sodium intake of 2500 mg/day and is smoking. What is your estimate of this student's weight?

(4 marks)

**Question 4c**

Interpret the adjusted coefficient of multiple determination. Explain the implication in this case is.

(6 marks)

**Question 4d**

Execute appropriate hypothesis tests to advise Mr Tan on which variable(s) is/are statistically significant at the 5% significance level. Write down your steps in detail.

(10 marks)

**----- END OF PAPER -----**